

Stochastic Differential Equation

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1 Prove of Ito's lemma

Given

$$dX(t) = f(t, X(t))dt + g(t, X(t))dW(t) \quad (1)$$

With another process $Y(t)$ has

$$Y(t) = \phi(t, X(t)) \quad (2)$$

Apply Taylor's expansion upon $Y(t)$

$$dY(t) = \phi_t(t, X(t))dt + \frac{1}{2}\phi_{tt}(t, X(t))dt^2 + \phi_X(t, X(t))dX + \frac{1}{2}\phi_{XX}(t, X(t))dX^2 \quad (3)$$

Bring equation (1) into equation (3), have

$$dY(t) = \phi_t(t, X(t))dt + \frac{1}{2}\phi_{tt}(t, X(t))dt^2 \quad (4)$$

$$+ \phi_X(t, X(t))[f(t, X(t))dt + g(t, X(t))dW(t)] \quad (5)$$

$$+ \frac{1}{2}\phi_{XX}(t, X(t))[f(t, X(t))dt + g(t, X(t))dW(t)]^2 \quad (6)$$

Since

$$[f(t, X(t))dt + g(t, X(t))dW(t)]^2 = f^2(t, X(t))dt^2 + g^2(t, X(t))dW^2(t) \quad (7)$$

$$+ 2f(t, X(t))dtg(t, X(t))dW(t) \quad (8)$$

$$= g^2(t, X(t))dt \quad (9)$$

The equation (4) to (6) can be written as

$$dY(t) = \phi_t(t, X(t))dt + \frac{1}{2}\phi_{tt}(t, X(t))dt^2 \quad (10)$$

$$+ \phi_X(t, X(t))[f(t, X(t))dt + g(t, X(t))dW(t)] \quad (11)$$

$$+ \frac{1}{2}\phi_{XX}(t, X(t))g^2(t, X(t))dt \quad (12)$$

Restructure the upon formula

$$dY(t) = [\phi_t(t, X(t)) + \phi_X(t, X(t))f(t, X(t)) + \frac{1}{2}\phi_{XX}(t, X(t))g^2(t, X(t))]dt \quad (13)$$

$$+ \phi_X(t, X(t))g(t, X(t))dW(t) \quad (14)$$

Say, if we assume

$$\phi(t, X(t)) = X^2 \quad (15)$$

With SDE of $dX(t) = dW(t)$, which means $f = 0$, and $g = 1$, then

$$dY(t) = [\phi_t(t, X(t)) + \frac{1}{2}\phi_{XX}(t, X(t))]dt \quad (16)$$

$$+ \phi_X(t, X(t))dW(t) \quad (17)$$

Given

$$\phi_X(t, X(t)) = 2X \quad (18)$$

and

$$\phi_t(t, X(t)) = 0 \quad (19)$$

and

$$\phi_{XX}(t, X(t)) = 2 \quad (20)$$

We have

$$dY(t) = dX^2(t) = dW^2(t) = dt + 2XdW(t) = dt + 2W(t)dW(t) \quad (21)$$

Integrate upon formula

$$W^2(t) = \int_0^t [dt + 2W(t)dW(t)] \quad (22)$$

$$= t + \int_0^t [2W(t)dW(t)] \quad (23)$$

Where

$$\int_0^t [W(t)dW(t)] = \frac{1}{2}W^2(t) - \frac{1}{2}t^2 \quad (24)$$

If we write Ito's lemma in vector notation, have

$$dY(t) = \tilde{f}(t, X(t))dt + \tilde{g}(t, X(t))dW(t) \quad (25)$$

$$\tilde{f}(t, X(t)) = \phi_t(t, X(t)) + \phi_X(t, X(t))f(t, X(t)) \quad (26)$$

$$+ \frac{1}{2}tr[\phi_{XX}(t, X(t))g(t, X(t))g^T(t, X(t))] \quad (27)$$

$$\tilde{g}(t, X(t)) = \phi_X(t, X(t))g(t, X(t)) \quad (28)$$

Assume the stochastic process U is combination of two stochastic processes X_1 and X_2 , like

$$U(t) = X_1(t)X_2(t) \quad (29)$$

With

$$dX_1(t) = f_1(t, X(t))dt + g_1(t, X(t))dW(t) \quad (30)$$

$$dX_2(t) = f_2(t, X(t))dt + g_2(t, X(t))dW(t) \quad (31)$$

Apply Ito's lemma, have

$$dU(t) = [X_2(t)f_1(t, X(t)) + X_1(t)f_2(t, X(t)) + g_1(t, X(t))g_2(t, X(t))]dt \quad (32)$$

$$+ [X_2(t)g_1(t, X(t)) + X_1(t)g_2(t, X(t))]dW(t) \quad (33)$$

Partial derivatives of U

$$\frac{\partial U}{\partial X} = (X_2(t), X_1(t))^T \quad (34)$$

$$\frac{\partial^2 U}{\partial X^2} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad (35)$$

$$\frac{\partial U}{\partial t} = 0 \quad (36)$$

Based on vector notation of Ito's lemma

$$dU(t) = \tilde{f}(t, X(t))dt + \tilde{g}(t, X(t))dW(t) \quad (37)$$

$$\tilde{f}(t, X(t)) = \frac{\partial U}{\partial t} + \frac{\partial U}{\partial X(t)} f(t, X(t)) \quad (38)$$

$$+ \frac{1}{2} tr \left[\frac{\partial^2 U}{\partial X^2} g(t, X(t)) g^T(t, X(t)) \right] \quad (39)$$

$$= \frac{\partial U}{\partial X(t)} f(t, X(t)) + \frac{1}{2} tr \left[\frac{\partial^2 U}{\partial X^2} g(t, X(t)) g^T(t, X(t)) \right] \quad (40)$$

$$= \begin{bmatrix} X_2(t) \\ X_1(t) \end{bmatrix} \begin{bmatrix} f_1(t, X(t)) \\ f_2(t, X(t)) \end{bmatrix} + \frac{1}{2} tr \left[\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} g_1(t, X(t)) \\ g_2(t, X(t)) \end{bmatrix} \begin{bmatrix} g_1(t, X(t)) & g_2(t, X(t)) \end{bmatrix} \right] \quad (41)$$

$$= \begin{bmatrix} X_2(t) \\ X_1(t) \end{bmatrix} \begin{bmatrix} f_1(t, X(t)) \\ f_2(t, X(t)) \end{bmatrix} + \frac{1}{2} tr \left[\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} g_1^2(t, X(t)) & g_1(t, X(t))g_2(t, X(t)) \\ g_2(t, X(t))g_1(t, X(t)) & g_2^2(t, X(t)) \end{bmatrix} \right] \quad (42)$$

$$= X_2(t)f_1(t, X(t)) + X_1(t)f_2(t, X(t)) + \frac{1}{2} [g_1(t, X(t))g_2(t, X(t)) + g_2(t, X(t))g_1(t, X(t))] \quad (43)$$

$$= X_2(t)f_1(t, X(t)) + X_1(t)f_2(t, X(t)) + g_1(t, X(t))g_2(t, X(t)) \quad (44)$$

$$\tilde{g}(t, X(t)) = \frac{\partial U}{\partial X(t)} g(t, X(t)) \quad (45)$$

$$= \begin{bmatrix} X_2(t) \\ X_1(t) \end{bmatrix} \begin{bmatrix} g_1(t, X(t)) \\ g_2(t, X(t)) \end{bmatrix} \quad (46)$$

$$= X_2(t)g_1(t, X(t)) + X_1(t)g_2(t, X(t)) \quad (47)$$

$$dU(t) = [X_2(t)f_1(t, X(t)) + X_1(t)f_2(t, X(t)) + g_1(t, X(t))g_2(t, X(t))]dt \quad (48)$$

$$+ [X_2(t)g_1(t, X(t)) + X_1(t)g_2(t, X(t))]dW(t) \quad (49)$$

If f and g are affine functions of one dimensional stochastic process X , then dX is linear or scalar SDE, like

$$f(t, X(t)) = A(t)X(t) + a(t) \quad (50)$$

$$g(t, X(t)) = \sum_{i=1}^m (B_i(t)X(t) + b_i(t)) \quad (51)$$

In general, X has solution

$$X(t) = \phi(t)(X(0) + \int_0^t \phi^{-1}(s)[a(s) - \sum_{i=1}^m B_i(s)b_i(s)]ds) \quad (52)$$

$$+ \sum_{i=1}^m \int_0^t \phi^{-1}(s)b_i(s)dW_i(s) \quad (53)$$

Phi is fundamental matrix, where

$$\phi(t) = \exp\left(\int_0^t \left[A(s) - \sum_{i=1}^m \frac{B_i^2(s)}{2}\right] ds + \sum_{i=1}^m \int_0^t B_i(s) dW_i(s)\right) \quad (54)$$

Define X first moment, mean, as m and second moment, variance, as P, have

$$\dot{m}(t) = A(t)m(t) + a(t), m(0) = X(0) \quad (55)$$

$$\dot{P}(t) = (2A(t) + \sum_{i=1}^m B_i^2(t))P(t) + 2m(t)(a(t) + \sum_{i=1}^m B_i(t)b_i(t)) + \sum_{i=1}^m b_i^2(t) \quad (56)$$

Now, here is prove

$$dm = E(dX) = E\left((A(t)X(t) + a(t))dt + \left(\sum_{i=1}^m (B_i(t)X(t) + b_i(t))\right)dW_i(t)\right) \quad (57)$$

$$= (A(t)E(X(t)) + a(t))dt + \left(\sum_{i=1}^m (B_i(t)E(X(t) + b_i(t)))E(dW_i(t))\right) \quad (58)$$

$$= (A(t)m(t) + a(t))dt \quad (59)$$

$$\dot{m}(t) = \frac{dm}{dt} = (A(t)m(t) + a(t)) \quad (60)$$

And assume $Y = X^2$, based on Ito's lemma

$$dY = \tilde{f}(t, X(t))dt + \tilde{g}(t, X(t))dW(t) \quad (61)$$

$$= \left(\frac{\partial Y}{\partial t} + \frac{\partial Y}{\partial X(t)}f(t, X(t)) + \frac{1}{2}\frac{\partial^2 Y}{\partial X^2(t)}g^2(t, X(t))\right)dt + \frac{\partial Y}{\partial X(t)}g(t, X(t))dW(t) \quad (62)$$

$$= (2X(t)f(t, X(t)) + \frac{1}{2}2g^2(t, X(t)))dt + 2X(t)g(t, X(t))dW(t) \quad (63)$$

$$= (2X(t)f(t, X(t)) + g^2(t, X(t)))dt + 2X(t)g(t, X(t))dW(t) \quad (64)$$

$$= (2X(t)(A(t)X(t) + a(t)) + \sum_{i=1}^m (B_i(t)X(t) + b_i(t))^2)dt \quad (65)$$

$$+ 2X(t)\left(\sum_{i=1}^m (B_i(t)X(t) + b_i(t))\right)dW(t) \quad (66)$$

$$= (2X(t)(A(t)X(t) + a(t)) + \sum_{i=1}^m (B_i^2(t)X^2(t) + b_i^2(t) + 2B_i(t)X(t)b_i(t)))dt \quad (67)$$

$$+ 2X(t)\left(\sum_{i=1}^m (B_i(t)X(t) + b_i(t))\right)dW(t) \quad (68)$$

$$(69)$$

Thus

$$dP = E(dY) = E((2X(t)(A(t)X(t) + a(t)) + \sum_{i=1}^m (B_i^2(t)X^2(t) + b_i^2(t) + 2B_i(t)X(t)b_i(t)))dt \quad (70)$$

$$+ E(2X(t)(\sum_{i=1}^m (B_i(t)X(t) + b_i(t)))dW(t)) \quad (71)$$

$$= E(2A(t)X^2(t) + 2a(t)X(t) + \sum_{i=1}^m (B_i^2(t)X^2(t) + b_i^2(t) + 2B_i(t)X(t)b_i(t)))dt \quad (72)$$

$$= (2A(t)E(X^2(t)) + 2a(t)E(X(t)) + \sum_{i=1}^m (B_i^2(t)E(X^2(t)) + b_i^2(t) + 2B_i(t)E(X(t))b_i(t)))dt \quad (73)$$

$$= (2A(t)P(t) + 2a(t)m(t) + \sum_{i=1}^m (B_i^2(t)P(t) + b_i^2(t) + 2B_i(t)m(t)b_i(t)))dt \quad (74)$$

$$\dot{P}(t) = \frac{dP}{dt} = (2A(t)P(t) + 2a(t)m(t) + \sum_{i=1}^m (B_i^2(t)P(t) + b_i^2(t) + 2B_i(t)m(t)b_i(t))) \quad (75)$$

Solve the above ODE can give the expected $N(m(t), P(t))$ of $P(X|X(0) = x_0)$
Here we begin to solve vector valued SDE

$$dX(t) = (A(t)m(t) + a(t))dt + (\sum_{i=1}^m (B_i(t)X(t) + b_i(t)))dW(t) \quad (76)$$

In which, $A(t), B_i(t) \in R^{n \times n}$, $a_i(t), b_i(t) \in R^n$, $W(t) \in R^m$ Linear SDE has solution like

$$X(t) = \phi(t)(X(0) + \int_0^t \phi^{-1}(t)[a(t) - \sum_{i=1}^m B_i(t)b_i(t)]dt \quad (77)$$

$$+ \sum_{i=1}^m \int_0^t \phi^{-1}(t)b_i(t)dW_i(t)) \quad (78)$$

With fundamental function Phi as homogenous stochastic differential equation, we get vector valued SDE solution

$$d\phi(t) = A(t)\phi(t)dt + \sum_{i=1}^m B_i(t)\phi(t)dW_i(t) \quad (79)$$

With $\phi(0) = I, I \in R^{n \times n}$ Rename latter part as dY , then

$$X(t) = \phi(t)(X(0) + \int_0^t \phi^{-1}(t)dY(t)) \quad (80)$$

$$dY(t) = [a(t) - \sum_{i=1}^m B_i(t)b_i(t)]dt + \sum_{i=1}^m b_i(t)dW_i(t) \quad (81)$$

Further simplify the upon formula

$$X(t) = \phi(t)Z(t) \quad (82)$$

$$Z(t) = X(0) + \int_0^t \phi^{-1}(t)dY(t) \quad (83)$$

$$dZ(t) = \phi^{-1}(t)dY(t) \quad (84)$$

$$= \phi^{-1}(t)[[a(t) - \sum_{i=1}^m B_i(t)b_i(t)]dt + \sum_{i=1}^m b_i(t)dW_i(t)] \quad (85)$$

$$= \phi^{-1}(t)[a(t) - \sum_{i=1}^m B_i(t)b_i(t)]dt + \phi^{-1}(s) \sum_{i=1}^m b_i(t)dW_i(t) \quad (86)$$

Apply vector valued Ito's lemma upon $X(t) = \phi(t)Z(t)$

$$dX = Z(t)d\phi(t) + \phi(t)dZ(t) + g_\phi(t)g_Z(t)dt \quad (87)$$

$$= Z(t)(A(t)\phi(t)dt + \sum_{i=1}^m B_i(t)\phi(t)dW_i(t)) + dY(t) + \sum_{i=1}^m B_i(t)\phi(t)\phi^{-1}(s) \sum_{i=1}^m b_i(s)dt \quad (88)$$

$$= Z(t)A(t)\phi(t)dt + \sum_{i=1}^m Z(t)B_i(t)\phi(t)dW_i(t) + dY(t) + \sum_{i=1}^m B_i(t) \sum_{i=1}^m b_i(s)dt \quad (89)$$

$$= (Z(t)A(t)\phi(t) + \sum_{i=1}^m B_i(t) \sum_{i=1}^m b_i(s) + a(t) - \sum_{i=1}^m B_i(t)b_i(t))dt + (\sum_{i=1}^m Z(t)B_i(t)\phi(t) + \sum_{i=1}^m b_i(t))dW_i(t) \quad (90)$$

$$= [X(t)A(t) + a(t)]dt + [\sum_{i=1}^m (X(t)B_i(t) + b_i(t))]dW_i(t) \quad (91)$$

The Phi function satisfies form of vector valued SDE, thus the Phi is solution of it Let m and P represents first and second moment of X Start with first moment

$$dm(t) = dE(X(t)) = E(d(X(t))) = E[X(t)A(t) + a(t)]dt \quad (92)$$

$$= [E(X(t))A(t) + E(a(t))]dt \quad (93)$$

$$= [m(t)A(t) + a(t)]dt \quad (94)$$

$$\dot{m} = m(t)A(t) + a(t) \quad (95)$$

About second moment

$$dP(t) = dE(X(t)X^T(t)) = Ed(X(t)X^T(t)) \quad (96)$$

$$d(X(t)X^T(t)) = [X^T(t)(A(t)X(t) + a(t)) + X(t)(A^T(t)X^T(t) + a^T(t)) \quad (97)$$

$$+ \sum_{i=1}^m (B_i(t)X(t) + b_i(t)) \sum_{i=1}^m (B_i^T(t)X^T(t) + b_i^T(t))]dt \quad (98)$$

$$+ [X^T(t) \sum_{i=1}^m (B_i(t)X(t) + b_i(t)) + X(t) \sum_{i=1}^m (B_i^T(t)X^T(t) + b_i^T(t))]dW(t) \quad (99)$$

$$drift = A(t)X(t)X^T(t) + a(t)X^T(t) + A^T(t)X^T(t)X(t) + a^T(t)X(t) \quad (100)$$

$$+ (\sum_{i=1}^m B_i(t)X(t) + \sum_{i=1}^m b_i(t))(\sum_{i=1}^m B_i^T(t)X^T(t) + \sum_{i=1}^m b_i^T(t)) \quad (101)$$

$$= A(t)X(t)X^T(t) + a(t)X^T(t) + A^T(t)X^T(t)X(t) + a^T(t)X(t) \quad (102)$$

$$+ \sum_{i=1}^m B_i(t)X(t)B_i^T(t)X^T(t) + \sum_{i=1}^m B_i(t)X(t)b_i^T(t) + \sum_{i=1}^m B_i^T(t)X^T(t)b_i(t) + \sum_{i=1}^m b_i(t)b_i^T(t) \quad (103)$$

$$(104)$$

dW has expectation of zero, thus the dP can be written as

$$dP(t) = E(A(t)X(t)X^T(t) + a(t)X^T(t) + A^T(t)X^T(t)X(t) + a^T(t)X(t)) \quad (105)$$

$$+ \sum_{i=1}^m (B_i(t)X(t)B_i^T(t)X^T(t) + B_i(t)X(t)b_i^T(t) + B_i^T(t)X^T(t)b_i(t) + b_i(t)b_i^T(t))dt \quad (106)$$

$$= A(t)P(t) + a(t)m^T(t) + A^T(t)P(t) + a^T(t)m(t) \quad (107)$$

$$+ \sum_{i=1}^m (B_i(t)B_i^T(t)P(t) + B_i(t)m(t)b_i^T(t) + B_i^T(t)m^T(t)b_i(t) + b_i(t)b_i^T(t))dt \quad (108)$$

$$\dot{P} = A(t)P(t) + a(t)m^T(t) + A^T(t)P(t) + a^T(t)m(t) \quad (109)$$

$$+ \sum_{i=1}^m (B_i(t)B_i^T(t)P(t) + B_i(t)m(t)b_i^T(t) + B_i^T(t)m^T(t)b_i(t) + b_i(t)b_i^T(t)) \quad (110)$$

Based on variance matrix definition, have

$$V = E(X(t)X^T(t)) - E(X(t))E(X^T(t)) = P(t) - m(t)m^T(t) \quad (111)$$

The above prove gives the n-dimensional stochastic motion Heston model is two-dimensional stochastic process, with X1 as asset price and X2 as volatility Here we begin to discuss non-linear, like

$$dX(t) = 1dt + 2X^{\frac{1}{2}}(t)dW(t) \quad (112)$$

$$(113)$$

Let $X(t) = \Phi(t) = (Y(t) + X(0))^2$, and, $dY(t) = dW(t)$

$$\Phi_Y(t) = 2(Y(t) + X(0)) \quad (114)$$

$$\Phi_{YY}(t) = 2 \quad (115)$$

$$d\Phi(t, Y(t)) = [\phi_t(t, Y(t)) + \phi_Y(t, Y(t))f(t, Y(t)) + \frac{1}{2}\phi_{YY}(t, Y(t))g^2(t, Y(t))]dt \quad (116)$$

$$+ \phi_Y(t, Y(t))g(t, Y(t))dW(t) \quad (117)$$

$$= (0 + 2(Y(t) + X(0))0 + \frac{1}{2}2)dt + 2(Y(t) + X(0))dW(t) \quad (118)$$

$$= 1dt + 2(Y(t) + X(0))dW(t) \quad (119)$$

$$dX(t) = 1dt + 2(X(t))^{\frac{1}{2}}dW(t) \quad (120)$$

In general, X can be expressed as

$$X = X(0) + \int_0^t f(s, X(s))ds + \int_0^t g(s, X(s))dW(s) \quad (121)$$

Drift integral is path-wise Riemann integral, and diffusion integral is Ito integral